pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Dates and Coupon Schedules

WEEK 2 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Week 2 settles between coupons

Settlement: Monday 30 November 2026

The same four bonds as week 1, two months after issue, between two coupons. Three questions come before any price:

1. **The previous coupon: interest accrues from here**
2. **The next coupon**
3. **The coupons left to be paid**

Days 7 to 9 build accrued interest, price, and duration on these three answers.

Week 2: dates, accrued, and risk

Day 6	**Dates and coupon schedules**
Day 7	Day counts and accrued interest
Day 8	Clean and dirty price
Day 9	Duration and DV01
Day 10	Convexity, the September book

Dates in Python, serial numbers in Excel

```
from datetime import date, timedelta
settlement = date(2026, 11, 30)
issue = date(2026, 9, 30)
(settlement - issue).days        # 61
settlement + timedelta(days=1)   # 2026-12-01
settlement < date(2031, 9, 30)   # True
```

A date holds a year, a month, and a day. One date minus another is a timedelta; dates compare in calendar order.

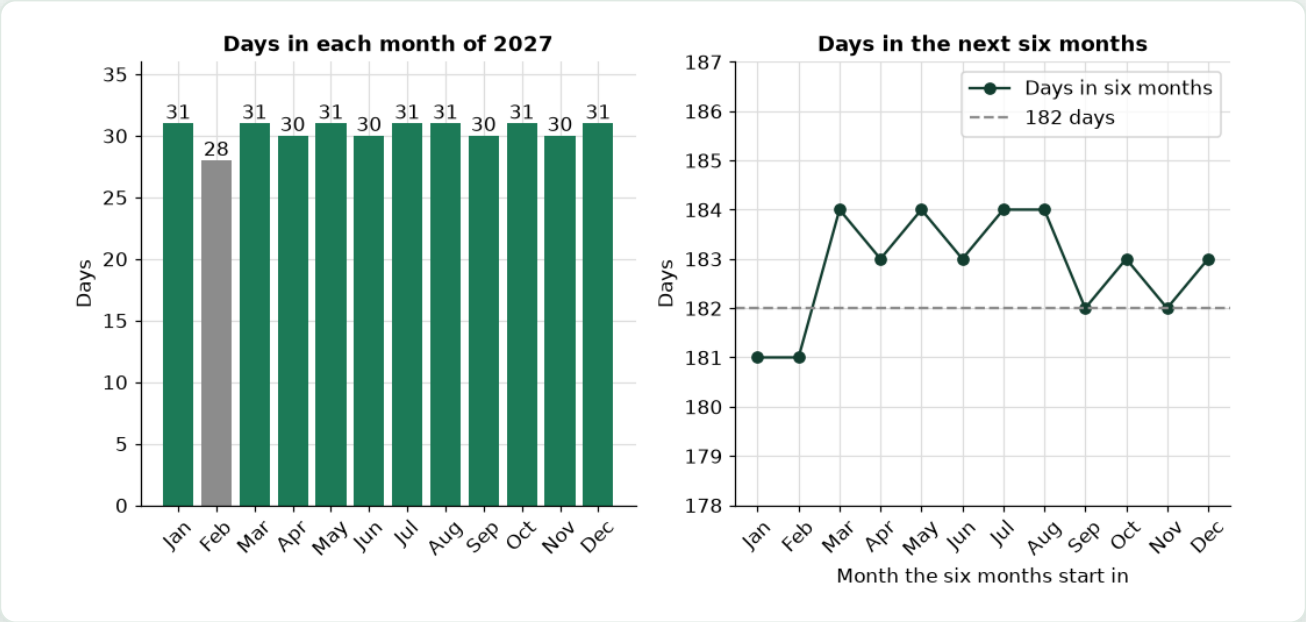
Excel side by side (tested in Excel)

`=DATE(2026,11,30), General`	46356
`=DATE(2026,9,30), General`	46295
`=DATE(2026,11,30)-DATE(2026,9,30)`	61
`=TEXT(DATE(2026,11,30),"dddd")`	Monday

Excel keeps a date as a count of days, and shows it in a date format.

Excel treats 1900 as a leap year, for Lotus 1-2-3 compatibility (Microsoft Learn). The story is in the notebook.

Adding months is not adding days



Six months is 181 to 184 days. Adding 182 days to 30 Sep 2026 and again gives 29 Sep 2027, not the coupon date.

date(2026, 9 + 6, 30) and date(2027, 2, 31) both raise ValueError: Python refuses a month 15 and a 31 February.

Moving a date by months needs two things

Carry the months into the year

Bring the day down in a short month

calendar.monthrange(year, month)[1] is the number of days in any month: 28 in February 2027, 29 in February 2028.

add_months: carry the year, clamp the day

```
# months counted from year 0
month_index = d.year * 12 + (d.month - 1) + months
year, month_zero_based = divmod(month_index, 12)
month = month_zero_based + 1
days_in_month = calendar.monthrange(year, month)[1]
return date(year, month, min(d.day, days_in_month))
```

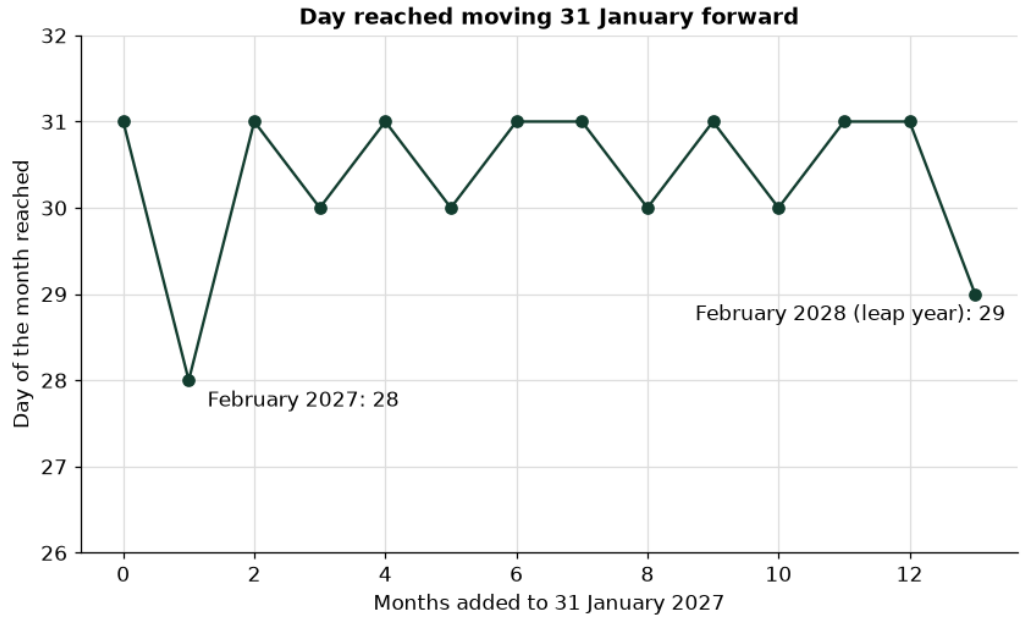
Date	Months	Result
2031-08-31	-6	2031-02-28
2032-08-31	-6	2032-02-29
2027-01-31	+1	2027-02-28
2031-09-30	-6	2031-03-30

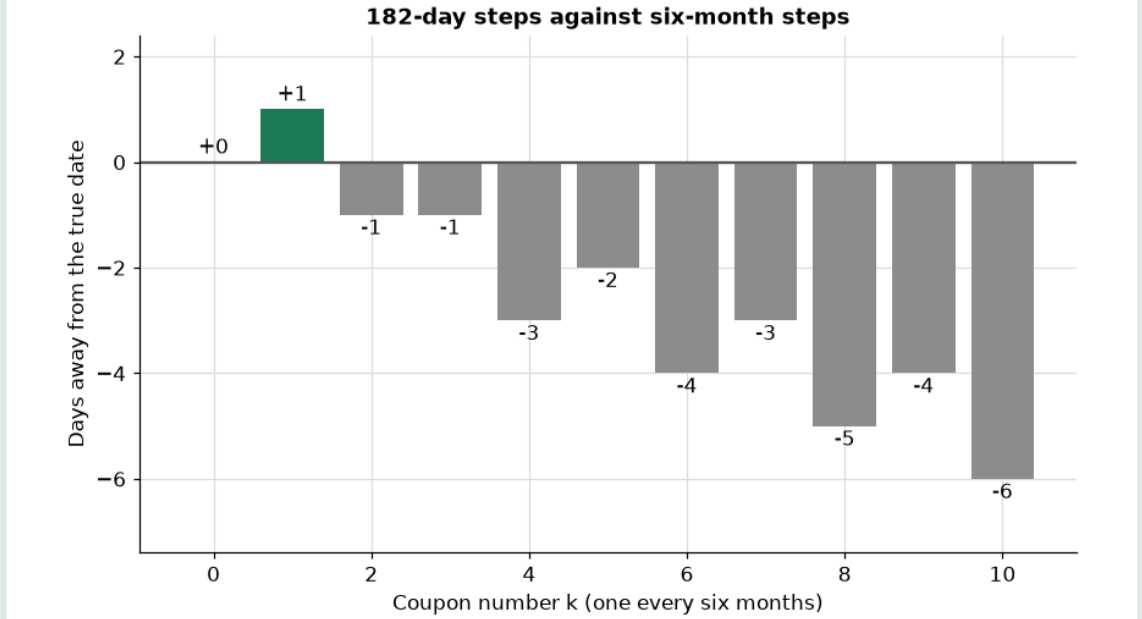
Excel side by side (tested in Excel)

`=EDATE(DATE(2031,8,31),-6)`	2031-02-28
`=EDATE(DATE(2032,8,31),-6)`	2032-02-29
`=EDATE(DATE(2027,1,31),1)`	2027-02-28
`=EDATE(DATE(2031,9,30),-6)`	2031-03-30

EDATE equals add_months on all five cases. Give the cell a date format to read it.

add_months, drawn

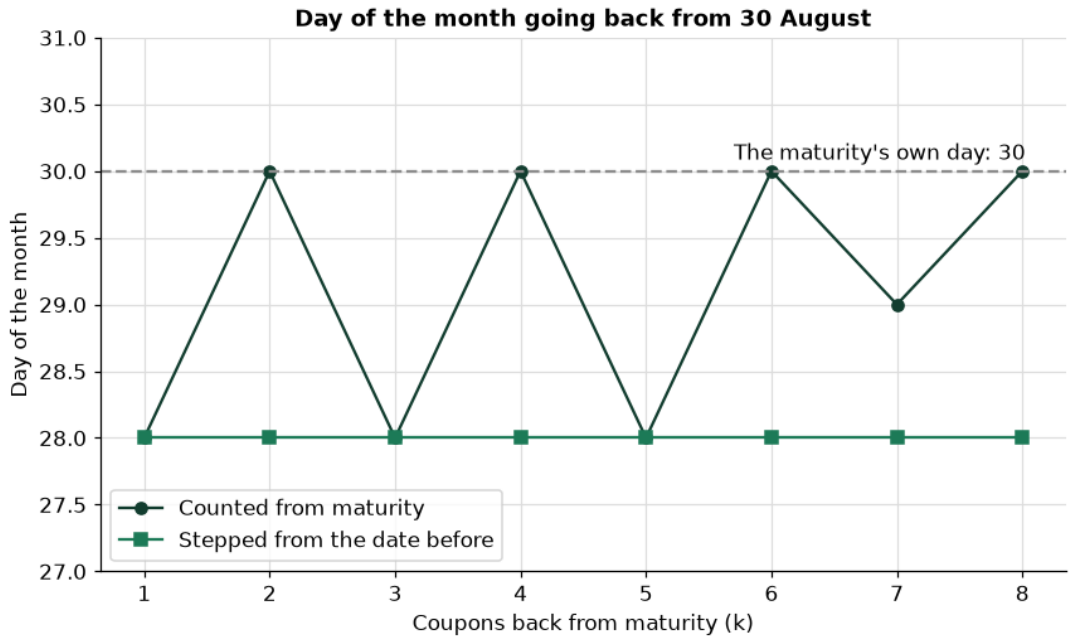




31 January moved on k months: the day comes down only as far as each month needs, then starts again from 31.

182-day steps wander from the true six-month date: 6 days early after five years.

Count every coupon date from maturity



Stepping carries February's 28th on. Counting from maturity goes back to the 30th.

Excel side by side (tested in Excel)

`A1: =DATE(2031,8,30)`	the maturity
`A2: =EDATE(A1,-6), filled down`	drifts to 28
`C2 down: 1, 2, 3, ...`	k
`B2: =EDATE($A$1,-6*C2), filled down`	keeps the 30th

EDATE matches Python. The drift comes from how the column is built.

The month-end rule

2027-03-30

add_months from Quorvane's maturity, 9 steps back

2027-03-31

Excel's COUPNCD on 30 November 2026

A bond that matures on the last day of a month pays every coupon on the last day of a month.

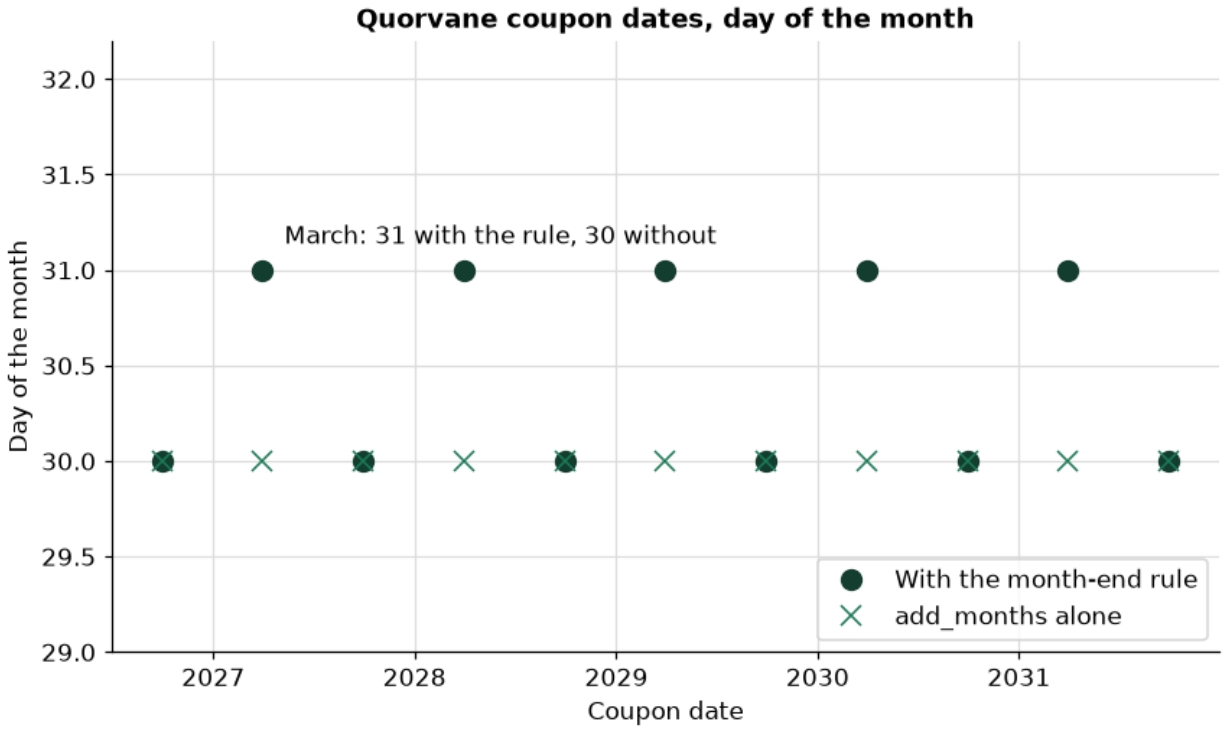
Otherwise the day is kept, and clamped in a short month. 30 September is a month end; 30 August is not; 28 February is in 2031 and not in 2032.

Excel side by side (tested in Excel)

`=EDATE(DATE(2031,9,30),-6)`	2031-03-30
`=EOMONTH(DATE(2031,9,30),-6)`	2031-03-31
`=EOMONTH(DATE(2031,9,30),-12)`	2030-09-30

EOMONTH moves by whole months to the last day of the month: where COUPNCD puts a month-end bond's coupons.

The month-end rule, drawn



Day of the month	March	September
With the month-end rule	31	30
add_months alone	30	30

In September both are on the 30th. In March the rule moves the coupon to the 31st, where Excel's COUPNCD puts it.

Previous coupon, next coupon, coupons left

Issuer	Previous	Next	Left
Quorvane Energy	2026-09-30	2027-03-31	10
Dunmarrow Rail	2026-09-30	2027-03-31	14
Pellwick Paper	2026-09-30	2027-03-31	6
Halvering Foods	2026-09-30	2027-03-31	20

Every value equals Excel's COUPPCD, COUPNCD, and COUPNUM.

Excel side by side (tested in Excel)

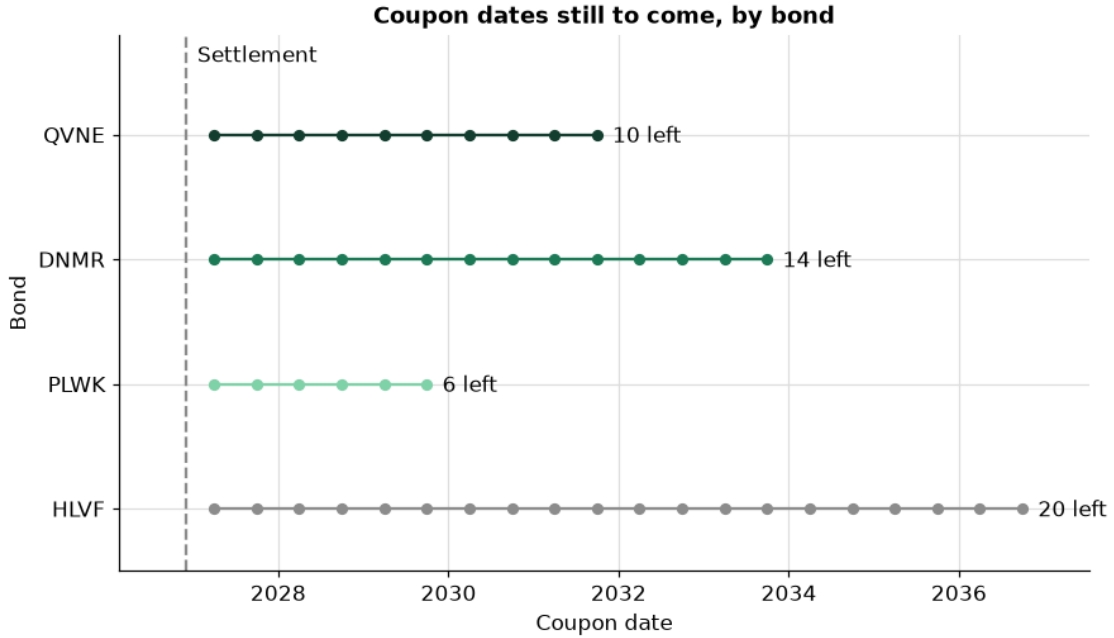
`=COUPPCD(DATE(2026,11,30),DATE(2031,9,30),2,0)`	2026-09-30
`=COUPNCD(DATE(2026,11,30),DATE(2031,9,30),2,0)`	2027-03-31
`=COUPNUM(DATE(2026,11,30),DATE(2031,9,30),2,0)`	10

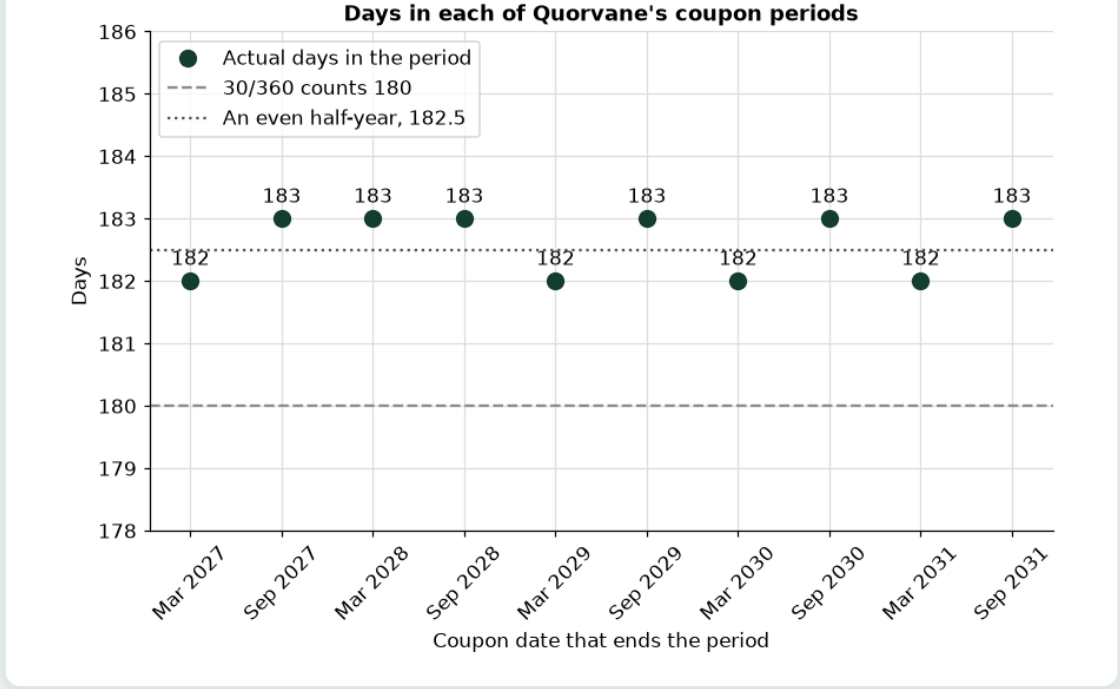
182 or 183

days in each of Quorvane's coupon periods, never 182.5

The previous coupon is the issue date: no coupon paid yet, but interest accrues from there.

The schedules, drawn





One row per bond, one dot per coupon still to come: 10, 14, 6, 20 dots, all starting on 31 March 2027.

No period is 182.5 days: each is 182 or 183, and 30/360 calls every one 180.

Date columns in pandas, and two systems

```
h["maturity"] = pd.to_datetime(h["maturity"])
h["maturity"].dt.year
h["maturity"].dt.is_month_end
h["maturity"].dt.date      # for bondmath
```

113

holdings bonds with a coupon before the 2026 year end

8,540,781.25

dollars of coupon on the par held

A description of the file's cash calendar, not of any bond's value.

Two systems disagree

Two benchmark bonds mature on 28 February in a non-leap year, a month end.

Convention	Previous coupon
Excel's month-end rule	31 Aug 2026
Data generator, day kept	28 Aug 2026

Both are conventions. A real bond's own documents decide.

bondmath follows Excel, so day 10 leaves these two out by name.

Today's Excel, every formula tested

Job	Excel (tested)	Returns
A date	`=DATE(2026,11,30), format General`	46356
Days between	`=DATE(2026,11,30)-DATE(2026,9,30)`	61
Move by months	`=EDATE(DATE(2031,8,31),-6)`	2031-02-28
To a month end	`=EOMONTH(DATE(2031,9,30),-6)`	2031-03-31
Previous coupon	`=COUPPCD(DATE(2026,11,30),DATE(2031,9,30),2,0)`	2026-09-30
Next coupon	`=COUPNCD(DATE(2026,11,30),DATE(2031,9,30),2,0)`	2027-03-31
Coupons left	`=COUPNUM(DATE(2026,11,30),DATE(2031,9,30),2,0)`	10

Excel returns a date as its serial number. Give the cell a date format to read it.

A .gitignore: keep cache folders out

git status --short, before

```
?? __pycache__/
?? bondmath_excel_reference.csv
?? ..._reference_tvm.csv
```

.gitignore

```
# folders Python and pytest make
__pycache__/
.pytest_cache/
```

git status --short, after

```
?? .gitignore
?? bondmath_excel_reference.csv
?? ..._reference_tvm.csv
```

__pycache__ holds compiled copies of the module: Python rebuilds it, and it differs by machine. The .gitignore itself is committed, so every copy ignores the same names.

First today's work is committed: Add add_months and coupon_dates, tested against Excel. Then the .gitignore.

Bring it home: paste, test, commit, ignore

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Today's two functions and two tests at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 15 passed, or 20 with the exercises of days 1 to 5.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Ignore

git status, Set-Content .gitignore, git status again, add, commit.

```
Set-Content .gitignore "__pycache__/", ".pytest_cache/"
```

In Colab: download bondmath.py and test_bondmath.py from the work folder. The file browser hides dot files, so make the .gitignore on your own machine.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert against Excel's reference rows. Run it: it fails.

4

Fix

Find the convention it broke, fix it, run the test again.

Today: each coupon date is stepped from the one before, so a 31 August maturity drifts to the 28th.

Code that runs is not the same as code that is right.

What you can do now

- Make, compare, and subtract dates; read Excel's serial numbers
- Move a date by months with add_months or EDATE
- Count coupon dates from maturity, and apply the month-end rule
- Find previous, next, and coupons left; match COUPPCD, COUPNCD, COUPNUM
- Use pandas date columns and .dt.date; keep caches out with .gitignore

Week 2, Day 2 Day counts and accrued interest: why 30/360 says 60 days where the calendar says 61